

Viability-Preserving Rollout Approximate Dynamic Programming for Required-Target Orbital Inspection

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Abstract— Autonomous orbital inspection must limit maneuver effort while completing prescribed observations under relative-motion and safety constraints. This paper develops a viability-preserving rollout approximate dynamic programming (ADP) formulation on a directed graph of audited orbital arcs. A finite Bellman prefix uses a deterministic task-aware base policy's completion cost as its approximate terminal value. Uncertified completions receive infinite value, while safe connecting actions remain available without immediate inspection gain. Under deterministic finite-graph assumptions, exhaustive rollout preserves completion and does not exceed the graph cost of a completing base policy. To separate planner failure from unavailable viewpoints, we independently certify route feasibility before evaluating the planners. In a development pilot, this defines a primary, feasibility-conditioned all-solvable benchmark of 59 cases from three 33-view graphs: depth-three ADP completes 59/59 missions, compared with 57/59 for one-step ADP and 33/59 for the task-aware greedy policy. Its mean graph-cost gap is 0.96% over 27 cases with a proven finite-graph optimum. Four additional shifted cases failing the visibility certificate are retained as a separate stress test. The original fixed-library confirmation is reported separately: depth-three ADP completes 43/50 test and 9/30

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Supplementary material containing data and code is available with this article. Color versions of one or more figures are available online at <https://ieeexplore.ieee.org>.

shifted missions, and reduces mean maneuver velocity increment, Δv , by 11.05% on 43 jointly completed test missions relative to one-step-ADP-seeded local search. These results support completion-certified sequencing while identifying viewpoint availability as a separate mission-level constraint.

Index Terms— spacecraft inspection, approximate dynamic programming, safe observable orbital arc, relative orbital dynamics, inspection requirements

I. Introduction

Orbital inspection provides visual evidence for assessing spacecraft condition and planning on-orbit servicing [1]. Robot-aided inspection and flaw detection were demonstrated during STS-85 [2], and subsequent planning studies address coverage of large structures with one or several free-flying inspectors [3]–[5]. An autonomous inspector must decide where to observe and how to reach each observation. Relative dynamics couple these decisions because viewpoint order changes transfer feasibility and cumulative maneuver effort.

Next-best-view methods model observation quality [6], kinodynamic planners combine coverage and differential constraints [3], [4], and relative-orbit planners exploit natural motion [7], [8]. A prescribed inspection mission also distinguishes mandatory surface targets from optional background coverage. An aggregate threshold can be reached while a required target remains unobserved. We therefore fix target identities and optimize the viewpoints and sequence needed to observe every prescribed item.

The planning challenge is to reduce cumulative maneuver effort while preserving a feasible continuation to all required observations. A low-cost immediate transfer can leave costly remaining observations or exhaust the budget before completing the task. Each safe observable orbital arc (SOOA) combines a source-dependent Hill–Clohessy–Wiltshire (HCW) transfer, a stabilized camera observation, and feasibility audits. Safe connecting arcs remain useful even when they reveal no new target immediately. Fig. 1 presents the resulting workflow, from mission inputs and arc construction to rollout ADP and inspection records.

The main contribution is a viability-preserving rollout approximate dynamic programming (ADP) formulation for maneuver-efficient required-target inspection. It specializes established rollout policy improvement [9], [10] using complete task-aware base-policy evaluations and finite-depth Bellman expansion. Actions are evaluated by approximate total graph cost, with infinite values rejecting uncertified completions. The analysis establishes completion preservation and base-policy no-degradation for SOOA sequencing, with exact dynamic programming providing a reference on tractable graphs.

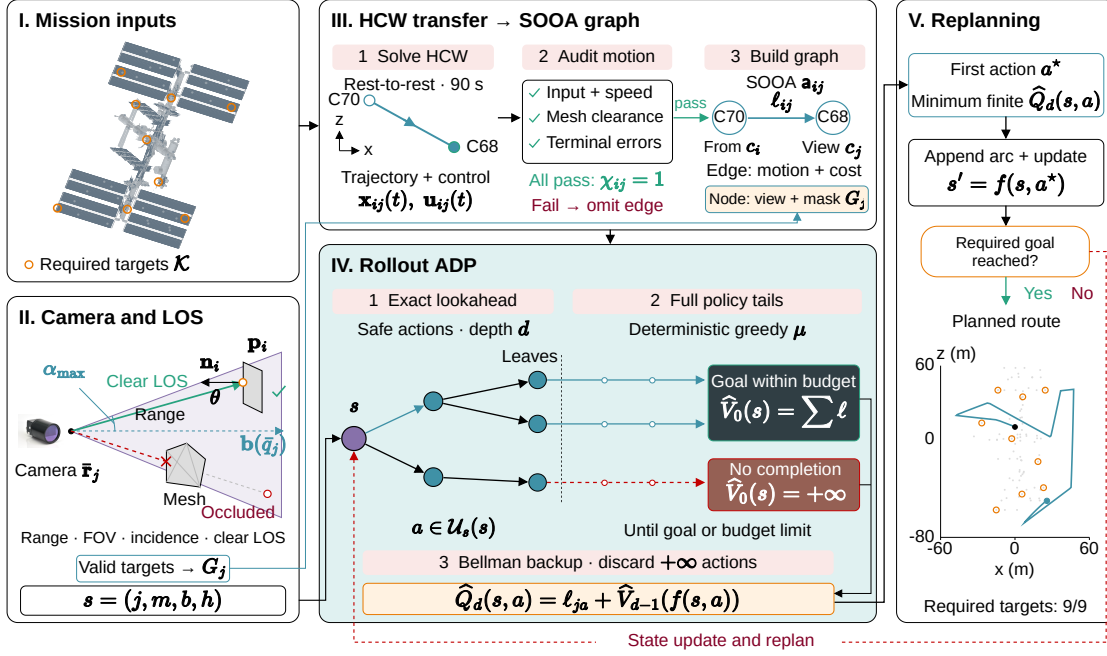


Fig. 1. Viability-preserving rollout ADP for orbital inspection. I. International Space Station (ISS) mesh and required targets $\mathcal{K}$. II. Camera-valid targets G_j from range, field of view (FOV), incidence $\theta \leq \theta_{\max}$, and line of sight (LOS) checks. III. HCW transfer, motion audits, and directed SOOA construction; C70 and C68 are archived candidate IDs. IV. Depth- d Bellman lookahead with complete greedy base-policy tails. At each leaf, $\sum \ell$ denotes the full tail cost; $+\infty$ denotes no certified completion within the remaining budget. V. First-action selection and planning-state updates toward completion of all required targets. The tree is schematic; the 9/9 route is an archived offline plan. ISS geometry: National Aeronautics and Space Administration (NASA), [NASA](https://www.nasa.gov).

The contributions are:

- 1) a viability-preserving rollout ADP over directed SOOA graphs for maneuver-efficient required-target inspection, combining required-target terminal predicates, finite-depth Bellman improvement, complete task-aware base-policy evaluation, audited visibility and motion transitions, and safe zero-gain connectors;
- 2) a finite-graph analysis establishing Markov sufficiency, specializing rollout completion and cost-improvement guarantees to SOOA sequencing, and proving exact optimality on exhausted reduced graphs; and
- 3) a reproducible evaluation of the method: an independently certified 59-case all-solvable benchmark (59/59 depth-three ADP versus 57/59 one-step ADP), a separate four-case visibility stress test, an 11.05% mean Δv reduction relative to one-step-ADP-seeded local search on 43 jointly completed test missions, and separate Robot Operating System 2 (ROS 2) execution checks.

II. Related Work

Orbital inspection and sensing: Inspection connects spacecraft observation to servicing decisions [1], [2]. Vision-based navigation has also been tested aboard the International Space Station (ISS) with the VER-TIGO/SPHERES facility [11]. That intravehicular micro-

gravity testbed provides experimental sensing evidence, distinct from external structure inspection. For external missions, camera visibility and collision avoidance must be considered together: Wang et al. combine position and attitude constraints in barrier-function-based inspection trajectory generation and tracking [12]. These requirements motivate the separation of observation acceptance, transfer admissibility, and task completion in OrbInspect.

Relative motion and inspection planning: The Clohessy–Wiltshire equations describe relative motion near a circular chief orbit [13], [14]. Williams derives inspection transfers using line-of-sight switching maneuvers [15]; Hou et al. design relative orbits for visual inspection of the China Space Station [7]; and Zhou et al. combine near-natural relative ellipses with formation control around a rotating station [16]. These approaches exploit orbital geometry and natural motion, which also determine the source-dependent costs and admissibility of the SOOA edges.

Robotic view planning couples sensor range, incidence, occlusion, and reconstruction quality to motion cost [6]. Rapidly exploring random trees and asymptotically optimal sampling methods provide motion-search foundations [17], [18]. Spacecraft inspection planners extend these ideas to kinodynamic coverage [3], [4], receding-horizon inspection of unknown noncooperative objects [19], and astrodynamics-informed transfer sampling [8]. Recent methods further address coverage–energy trade-offs [5] and surface-aware tomographic paths [20]. The present decision problem instead fixes a can-

didate library and prescribed target identities, then optimizes their completion sequence over audited transfers.

ADP and learning-based spacecraft control: ADP replaces expensive exact cost-to-go calculations with tractable approximations [10], [21]. Combinatorial rollout uses heuristic completions to improve sequential decisions [9]; its policy-iteration interpretation also connects rollout with model predictive control [22]. Value and policy iteration provide a broader theoretical basis for terminal-set optimal control [23]. OrbInspect uses a complete base-policy simulation as its value approximation, rather than a learned critic.

Reinforcement learning offers a complementary route to approximate optimal feedback control from trajectory data [24]. Spacecraft applications include constrained six-degree-of-freedom proximity control [25], online attitude stabilization under dynamic uncertainty [26], and actor-critic ADP with motion constraints, input saturation, and disturbance compensation [27]. Reach-avoid differential graphical games provide a related ADP setting for nonlinear multi-agent dynamics with communication constraints and input saturation [28]; their adversarial objectives differ from the deterministic required-observation sequencing studied here. For inspection specifically, van Wijk et al. combine deep reinforcement learning (DRL) with discrete control barrier functions and run-time assurance [29]. These studies concern learned control policies; the present model-based ADP contribution concerns completion-certified observation sequencing.

Safety and completion guarantees: Constrained policy optimization regulates expected constraint costs [30], whereas shielding restricts action selection using a specified safety model [31]. Control barrier functions impose conditions for forward invariance [32], and predictive barrier functions provide a recovery mechanism toward the feasible set of a predictive safety filter [33]. OrbInspect separates these control-level concerns from finite-task viability: motion audits admit SOOA edges, while finite rollout values certify a continuation to every required target. The resulting completion and cost bounds specialize established rollout principles to a directed inspection graph; they do not replace continuous-state safety certificates.

III. System and Observation Model

This section defines the deterministic system, observation, and feasibility model from which the candidate graph is constructed. The inspected structure is represented by a known station mesh and surface samples; the chaser state, camera parameters, nonempty required-target set, keep-out model, and actuation limits define each scenario. These ingredients determine visibility masks, admissible transfers, and SOOA edge costs.

The modeling assumptions and the resulting limits on interpretation are summarized in Table I before the system equations are introduced.

A. Relative Dynamics

The inspected structure is fixed in a local-vertical/local-horizontal (LVLH) frame attached to a nominal circular chief orbit. Its axes are radial, along-track, and cross-track, respectively. The chaser state is $\mathbf{x} = [\mathbf{r}^\top, \mathbf{v}^\top]^\top \in \mathbb{R}^6$, where $\mathbf{r} = [r_x, r_y, r_z]^\top$ and $\mathbf{v} = [v_x, v_y, v_z]^\top$. With mean motion n and commanded acceleration $\mathbf{u} = [a_x, a_y, a_z]^\top$, the HCW model is

$$\begin{aligned} \dot{\mathbf{x}} &= A_{\text{HCW}}\mathbf{x} + B\mathbf{u}, \\ A_{\text{HCW}} &= \begin{bmatrix} 0 & I \\ K_n & C_n \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ I \end{bmatrix}, \\ K_n &= \text{diag}(3n^2, 0, -n^2), \quad C_n = \begin{bmatrix} 0 & 2n & 0 \\ -2n & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned} \quad (1)$$

where I is the 3×3 identity matrix and the zero blocks have compatible dimensions. For the continuous HCW model, piecewise-constant acceleration over Δt has the exact zero-order-hold map

$$\begin{aligned} \mathbf{x}_{k+1} &= A_d \mathbf{x}_k + B_d \mathbf{u}_k, \\ A_d &= e^{A_{\text{HCW}} \Delta t}, \\ B_d &= \int_0^{\Delta t} e^{A_{\text{HCW}} \tau} B d\tau. \end{aligned} \quad (2)$$

B. Visibility, Prescribed Observation, and Coverage

The structure is represented by indexed surface samples $\{(\mathbf{p}_i, \mathbf{n}_i, a_i)\}_{i=1}^N$, where $\mathbf{p}_i \in \mathbb{R}^3$ is the sample position in the LVLH frame, $\mathbf{n}_i \in \mathbb{R}^3$ is its outward unit normal, and $a_i > 0$ is its represented surface area. Let $\mathcal{T} = \{1, \dots, N\}$ denote the sample-index set. Mission priority weights are separately defined as $w_i = a_i \eta_i$, where $\eta_i > 0$ is a priority multiplier; they are not geometric areas. A fixed nonempty set $\mathcal{K} \subseteq \mathcal{T}$ specifies the required inspection items before scenario evaluation. The camera attitude quaternion q_k defines unit boresight $\mathbf{b}(q_k)$, and $\mathcal{O}$ denotes the occluding mesh. Target index i is visible when

$$\begin{aligned} r_{\min}^{\text{cam}} &\leq \|\mathbf{p}_i - \mathbf{r}_k\| \leq r_{\max}^{\text{cam}}, \\ \arccos \frac{(\mathbf{p}_i - \mathbf{r}_k)^\top \mathbf{b}(q_k)}{\|\mathbf{p}_i - \mathbf{r}_k\|} &\leq \alpha_{\max}, \\ \arccos \frac{(\mathbf{r}_k - \mathbf{p}_i)^\top \mathbf{n}_i}{\|\mathbf{r}_k - \mathbf{p}_i\| \|\mathbf{n}_i\|} &\leq \theta_{\max}, \\ \text{LOS}(\mathbf{r}_k, \mathbf{p}_i, \mathcal{O}) &= 1. \end{aligned} \quad (3)$$

Here $r_{\min}^{\text{cam}}$ and $r_{\max}^{\text{cam}}$ are the camera range limits, $\alpha_{\max}$ is the FOV half-angle, $\theta_{\max}$ is the incidence-angle limit, and $\text{LOS}(\mathbf{r}_k, \mathbf{p}_i, \mathcal{O}) = 1$ denotes an unobstructed LOS. Let $\mathbf{m}_k \in \{0, 1\}^N$ denote the cumulative covered-target mask after the first k selected observations, initialized at $\mathbf{m}_0 = \mathbf{0}$. Let M denote the number of candidate observation nodes, indexed by $j = 1, \dots, M$. For a target-index set G_j , let $\mathbf{1}_{G_j} \in \{0, 1\}^N$ denote its indicator vector; let $\mathbf{e}_a \in \{0, 1\}^M$ denote the standard basis vector that marks selected destination a . Fig. 1, panel II,

TABLE I
Modeling assumptions and the boundary of the reported evidence

Assumption	Role in the baseline	Consequence for interpretation
HCW and fixed local-vertical/local-horizontal (LVLH) geometry	A circular chief orbit supports linear HCW relative translation, while a station fixed in LVLH keeps the mesh and safety geometry stationary during one planning run.	Quantitative results use a local linear, stationary-mesh model; perturbed ephemerides or station attitude motion require an extension.
Known mesh and state	Makes visibility, occlusion, and transfer evaluation deterministic.	Navigation, map, and shape uncertainty are not represented by the present safety margin.
Prescribed camera attitude	Points the boresight toward each candidate observation target and exports the corresponding quaternion stream.	Camera slew time, reaction-wheel limits, and closed-loop attitude error are outside the optimization.
Mesh-based sampled safety	Computes distance to the full International Space Station (ISS) triangle mesh at every rollout sample and rejects sampled motion segments that intersect the mesh.	Positive sampled clearance and surface nonintersection are planning audits rather than a robust continuous-time collision certificate.

illustrates the range, FOV, incidence-angle, and LOS tests in (3). LOS is queried against a deterministic bounding-volume hierarchy containing every ISS mesh triangle; intersections at the target endpoint are excluded, while interior segment intersections block the view. In a closed-loop execution model, dwell could be accumulated as $\tau_{i,k+1} = \tau_{i,k} + \Delta t$ while target i remains visible and accepted when $\tau_{i,k+1} \geq \tau_{\min}$, where $\tau_{i,k}$ is accumulated dwell time and $\tau_{\min}$ is the required dwell time. The offline study instead treats every candidate as a stabilized terminal observation with its prescribed exposure completed a priori. Applying (3) at the candidate pose $(\mathbf{r}_k, q_k) = (\bar{\mathbf{r}}_j, \bar{q}_j)$ yields its fixed visibility mask G_j . Selecting node j credits this mask, so $\mathbf{m}_{k+1} = \mathbf{m}_k \vee \mathbf{1}_{G_j}$. Here $\mathbf{m}_k = [m_{1,k}, \dots, m_{N,k}]^T$, where $m_{i,k} = 1$ exactly when target/sample i has been credited after the first k observations. Thus $\mathbf{m}_0 = \mathbf{0}$ is the initial all-uncovered mask, while $\mathbf{m}_k$ denotes the generic mask at decision step k . The sampled-area coverage represented by mask $\mathbf{m}_k$ is

$$C^{\text{area}}(\mathbf{m}_k) = \frac{\sum_i a_i m_{i,k}}{\sum_i a_i}. \quad (4)$$

Candidate generation can make a subset of the sampled surface invisible from every candidate observation. For the nonempty set $\mathcal{T}_{\text{insp}} = \cup_{j=1}^M G_j$, we also report priority-weighted inspectable coverage:

$$C_k^{\text{insp}} = \frac{\sum_{i \in \mathcal{T}_{\text{insp}}} w_i m_{i,k}}{\sum_{i \in \mathcal{T}_{\text{insp}}} w_i}. \quad (5)$$

Equation (4) estimates credited area over the original surface samples, whereas (5) is a priority-weighted score conditional on the candidate library. Their denominators remain fixed after scenario-specific observation loss. Neither is the primary required-target completion metric. Here, $\text{supp}(\mathbf{m}) = \{i \in \mathcal{T} : m_i = 1\}$ denotes the set of sample indices covered by binary mask $\mathbf{m}$.

$$C^{\mathcal{K}}(\mathbf{m}) = \frac{|\mathcal{K} \cap \text{supp}(\mathbf{m})|}{|\mathcal{K}|}, \quad (6)$$

$$\Gamma_{\mathcal{K}}(\mathbf{m}) = \mathbf{1}\{\mathcal{K} \subseteq \text{supp}(\mathbf{m})\}.$$

Mission success requires $\Gamma_{\mathcal{K}}(\mathbf{m}) = 1$, independently of optional area coverage. One observation can credit several

required items, and different viewpoints can satisfy the same item. Completion denotes acceptance of the prescribed samples under (3). Reported mission duration is transfer-only; observation exposure is assumed complete at each stabilized candidate.

C. Dynamic Feasibility and SOOAs

The finite-body radius is $r_{\text{veh}} = 0.80 \text{ m}$, which conservatively bounds the chaser visual model, and the prescribed mesh-clearance margin is $d_{\text{safe}} = 2.0 \text{ m}$. For an edge rollout with K post-step samples, let ε_r and ε_v denote the terminal position and velocity tolerances. Every sampled state must satisfy

$$\|\mathbf{u}_k\|_2 \leq u_{\max}, \quad \|\mathbf{v}_k\|_2 \leq v_{\max}, \quad d_S(\mathbf{r}_k) \geq 0, \quad (7)$$

where $d_S(\mathbf{r}) = d_{\text{mesh}}(\mathbf{r}) - r_{\text{veh}} - d_{\text{safe}}$ is finite-body clearance above the required margin, d_{mesh} is unsigned center distance to the closest triangle in the complete ISS mesh, and $d_S = 0$ is the sampled admissibility boundary. The mesh coordinates are obtained from the complete GL Transmission Format (glTF) scene hierarchy, followed by the Simulation Description Format (SDF) pose and scale. In addition, each consecutive pair of rollout samples is tested for a triangle-surface intersection, and its continuous center-distance lower bound subtracts half the segment length from the smaller endpoint distance by the 1-Lipschitz property of distance to a closed set. This swept test catches a direct mesh crossing even if both endpoints have positive unsigned distance. For a directed rollout from source node i to destination node j , define $\Delta v_{ij} = \sum_k \|\mathbf{u}_{i,j,k}\|_2 \Delta t$, root-mean-square distance e_{ij}^{rms} from the transfer samples to the destination position, and $d_{ij}^{\min} = \min_k d_S(\mathbf{r}_{i,j,k})$. The coefficients λ_v , λ_e , λ_s , and λ_a are fixed nonnegative weights for maneuver increment, terminal tracking error, clearance violation, and per-edge action cost, respectively. The graph stage cost is

$$\ell_{ij} = \lambda_v \Delta v_{ij} + \lambda_e e_{ij}^{\text{rms}} + \lambda_s \max(0, -d_{ij}^{\min}) + \lambda_a. \quad (8)$$

For any route, J denotes the sum of its edge stage costs; J_{rollout} in the illustrative figure is the corresponding

realized rollout cost. For every shield-admissible edge, the clearance-violation term is zero. It remains in the shared cost implementation so deliberately unshielded comparison methods can be evaluated with the same expression.

A candidate observation node and a directed SOOA edge are, respectively,

$$\begin{aligned} c_j &= (\bar{\mathbf{r}}_j, \bar{q}_j, G_j), \\ a_{ij} &= (\mathbf{x}_{ij}[1:K], \mathbf{u}_{ij}[0:K-1], q_{ij}[1:K], \\ &\quad \Delta v_{ij}, d_{ij}^{\min}, u_{ij}^{\max}, v_{ij}^{\max}, e_{ij}^r, e_{ij}^v, \rho_{ij}). \end{aligned} \quad (9)$$

Here $\bar{\mathbf{r}}_j$ and $\bar{q}_j$ are the terminal observation pose and prescribed attitude, $G_j \subseteq \mathcal{T}$ is its camera-valid target set, and e_{ij}^r and e_{ij}^v are terminal position and velocity errors. The stored state and attitude arrays contain the K post-step samples, while $u_{ij}^{\max}$ and $v_{ij}^{\max}$ record the maximum sampled input and speed norms; the source state is held by candidate node c_i . In the edge notation below, i and j index candidate nodes and are distinct from the surface-sample index used above. The initial condition is a distinguished source node c_0 with no visibility reward. For a passive audit with horizon $T_p = L_p \Delta t$, where L_p is the number of passive steps, zero-input HCW drift can be propagated from every stored edge state and recorded as $\rho_{ij} = \min_{1 \leq k \leq K, 0 \leq l \leq L_p} d_S(\mathbf{r}_{ij,k}^{\text{drift}}(l))$. The reported study disables this optional audit and therefore makes no passive-safety claim; it retains input, speed, full-mesh sampled clearance, swept surface-intersection, and terminal checks. Throughout this paper, “safe” means admissible under the audits explicitly enabled for the reported experiment.

The preceding model induces the required-target sequencing problem. The graph state and shield notation used to solve it are introduced in Section IV.

PROBLEM 1 (Required-target SOOA planning) *Given the candidate nodes and stored directed SOOA records above, an initial action budget H , and the required set $\mathcal{K}$, find a finite sequence $\pi = (a_1, \dots, a_L)$, with $0 \leq L \leq H$, that minimizes*

$$J(\pi) = \sum_{k=1}^L \ell_{j_{k-1}a_k}, \quad j_0 = 0, \quad j_k = a_k, \quad (10)$$

subject to the following conditions: each traversed record $a_{j_{k-1}a_k}$ passes all enabled motion, terminal, clearance, and passive audits; no destination is selected more than once; and the masks updated by $\mathbf{m}_k = \mathbf{m}_{k-1} \vee \mathbf{1}_{G_{a_k}}$, $\mathbf{m}_0 = \mathbf{0}$, satisfy $\Gamma_{\mathcal{K}}(\mathbf{m}_L) = 1$. A zero-gain action is allowed when it is needed to reach a later required observation. If no such sequence exists within H , the instance is declared infeasible in the available candidate graph.

IV. Viability-Preserving Rollout ADP

A. Candidate Graph and Markov State

Let $\mathcal{V} = \{c_0, c_1, \dots, c_M\}$ contain the initial node and the M candidate observation nodes in (9). A directed edge

from c_i to c_j is the source-dependent SOOA a_{ij} , and it is absent when an enabled audit fails. Thus visibility G_j is node dependent, whereas trajectory, cost, clearance, input, terminal error, and passive margin are edge dependent. This separation is necessary because reaching the same observation from two different source nodes generally produces different HCW motion and audit values.

A node identifier is not a sufficient decision state because the value of a view changes after its targets have been covered. The finite-horizon Markov state is

$$s_k = (j_k, \mathbf{m}_k, b_k, h_k), \quad (11)$$

where $j_k \in \{0, \dots, M\}$ is the current node, $\mathbf{m}_k \in \{0, 1\}^N$ is the covered-target mask, $b_k \in \{0, 1\}^M$ is the selected-observation mask, and h_k is the remaining action budget. The scalar destination index a is distinct from the surface-area scalar a_i defined above. Selecting destination index $a \in \{1, \dots, M\}$ traverses edge $a_{j_k a}$ and gives the deterministic transition

$$f(s_k, a) = (a, \mathbf{m}_k \vee \mathbf{1}_{G_a}, b_k \vee \mathbf{e}_a, h_k - 1). \quad (12)$$

The required set $\mathcal{K}$ is fixed problem data. The shield-admissible action set $\mathcal{U}_s(s)$ is defined below. The constrained graph problem minimizes total stage cost subject to (6), with zero terminal cost for a completed task and value $+\infty$ when no admissible completion exists within the remaining budget. Its Bellman recursion is

$$\begin{aligned} V^*(s) &= \begin{cases} 0, & \Gamma_{\mathcal{K}}(\mathbf{m}) = 1, \\ +\infty, & h = 0 \text{ or } \mathcal{U}_s(s) = \emptyset, \\ \min_{a \in \mathcal{U}_s(s)} Q^*(s, a), & \text{otherwise,} \end{cases} \\ Q^*(s, a) &= \ell_{ja} + V^*(f(s, a)). \end{aligned} \quad (13)$$

Here j and h are the current-node and remaining-budget components of s , and f is the deterministic state-transition map. Exact evaluation is exponential in the coverage and selected-node masks. The proposed method approximates this recursion with a short exact prefix and a computationally inexpensive, goal-aware terminal policy.

B. SOOA Construction and Safety Shield

Area-weighted targets are sampled from the mesh, and camera seeds are generated on offset shells around their outward normals. The binary visibility matrix $V_{ij} = \mathbf{1}\{i \in G_j\}$ is computed once from the terminal observation geometry. For a transition to $\bar{\mathbf{x}}_j = [\bar{\mathbf{r}}_j^\top, \mathbf{0}^\top]^\top$, the planner solves the minimum-energy piecewise-constant HCW boundary-value problem under the numerical fourth-order Runge–Kutta (RK4) map

$$\begin{aligned} \min_{\mathbf{u}_0, \dots, \mathbf{u}_{K-1}} \quad & \sum_{k=0}^{K-1} \|\mathbf{u}_k\|_2^2 \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = \tilde{A}_d \mathbf{x}_k + \tilde{B}_d \mathbf{u}_k, \\ & \mathbf{x}_K = \bar{\mathbf{x}}_j. \end{aligned} \quad (14)$$

If the terminal influence matrix $\mathcal{B}_K = [\tilde{A}_d^{K-1} \tilde{B}_d \quad \dots \quad \tilde{B}_d] \in \mathbb{R}^{6 \times 3K}$ has full row rank,

stack the commands as $\mathbf{U}_{ij} = [\mathbf{u}_{ij,0}^\top, \dots, \mathbf{u}_{ij,K-1}^\top]^\top$. The minimum-norm command from source state $\mathbf{x}_i^{\text{src}}$ is then $\mathbf{U}_{ij}^* = \mathcal{B}_K^\top (\mathcal{B}_K \mathcal{B}_K^\top)^{-1} (\bar{\mathbf{x}}_j - \tilde{A}_d^K \mathbf{x}_i^{\text{src}})$. The implementation computes this expression through the 6×6 Gram system and then propagates the requested input without clipping. Let $\chi_{ij} = 1$ exactly when every stored edge sample satisfies (7), the enabled swept-segment audit reports no triangle intersection, $e_{ij}^r \leq \varepsilon_r$, $e_{ij}^v \leq \varepsilon_v$, and $\rho_{ij} \geq 0$ when the passive audit is enabled. The shield then admits

$$\mathcal{U}_s(s) = \{a \in \{1, \dots, M\} : h > 0, b_a = 0, \chi_{ja} = 1\}, \quad (15)$$

where j is the current-node component of s . Safe unvisited nodes remain admissible when they provide no immediate inspection gain, because a connecting action may be needed to reach a required target on the directed graph. The finite budget and no-revisit mask bound such connections. The shield makes an inadmissible edge unavailable to incumbent construction, local search, and exact recursion [31]. Its guarantee remains conditional on the sampled numerical rollout and deterministic model used to compute χ_{ij} .

An edge is audit-admissible when $\chi_{ij} = 1$; an action is shield-admissible when it belongs to $\mathcal{U}_s(s)$. A scenario is feasible for the candidate graph when at least one shield-admissible sequence reaches $\Gamma_{\mathcal{K}}(\mathbf{m}) = 1$ within the available budget, and is infeasible when no such sequence exists.

C. Adaptive Base Policy and Rollout Value

Let $g_{\mathcal{K}}(s, a) = |\mathcal{K} \cap (G_a \setminus \text{supp}(\mathbf{m}))|$ count newly accepted required targets, and let $\epsilon = 0.05$ prevent division by a near-zero stage cost. For the required-target task, the deterministic adaptive base policy is

$$\mu(s) \in \arg \max_{a \in \mathcal{U}_s(s)} \left(\frac{g_{\mathcal{K}}(s, a)}{\max\{\epsilon, \ell_{ja}\}}, g_{\mathcal{K}}(s, a), -\ell_{ja} \right), \quad (16)$$

where the tuple is compared lexicographically, followed by deterministic node-identifier tie breaking. The policy recomputes gain after every accepted observation; zero-gain ties favor lower stage cost. Starting from state s , its completion value is

$$\hat{V}_0(s) = \begin{cases} \sum_{k=0}^{L-1} \ell_{jk} \mu(s_k), & \mu \text{ reaches } \Gamma_{\mathcal{K}}(\mathbf{m}) = 1 \text{ within } h, \\ +\infty, & \text{otherwise.} \end{cases} \quad (17)$$

where L is the number of actions taken by the base policy before completion or budget exhaustion. Thus $\hat{V}_0$ approximates the cost-to-go through a complete, shielded base-policy simulation.

For rollout depth $d \geq 1$, define the truncated Bellman value

$$\hat{V}_d(s) = \begin{cases} 0, & \Gamma_{\mathcal{K}}(\mathbf{m}) = 1, \\ +\infty, & h = 0, \\ \min_{a \in \mathcal{U}_s(s)} \hat{Q}_d(s, a), & \text{otherwise,} \end{cases} \quad (18)$$

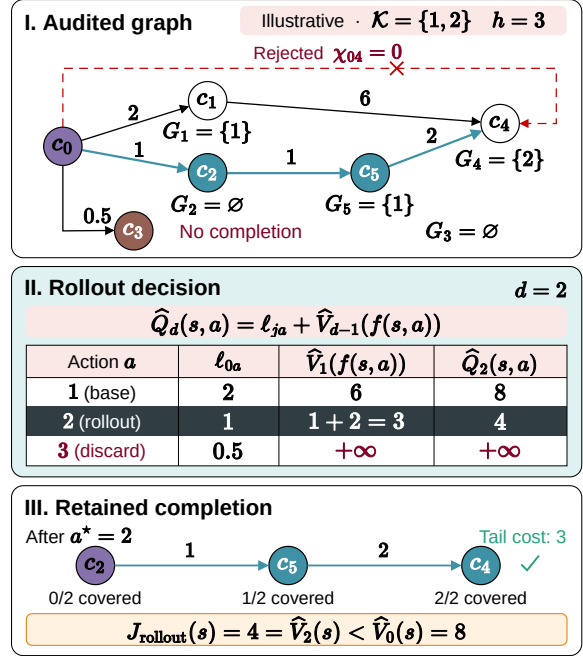


Fig. 2. Illustrative completion-certified rollout ADP, with initial state $s = (0, 0, 0, 3)$. I. Solid edges are admissible, edge labels are stage costs, and G_j lists visible targets. The dashed transfer is excluded. II. At $d = 2$, rollout selects zero-gain connector c_2 for total cost 4, versus base-policy cost 8; action 3 has no certified completion. III. The successor $s' = (2, 0, e_2, 2)$ retains a cost-3 completion. Counts indicate covered required targets; J_{rollout} is realized graph cost. The example illustrates Propositions 4 and 5. Assigned costs and audit flags are illustrative, not experimental HCW data; the primary study uses $d = 3$.

where $\hat{Q}_d(s, a) = \ell_{ja} + \hat{V}_{d-1}(f(s, a))$. The minimum over an empty action set is $+\infty$. The policy selects a finite-valued minimizing action, applies (12), and rebuilds the depth- d tree at the successor state. State-depth memoization removes duplicate subproblems within each decision. The guarantees below require exhaustive prefix evaluation and the same deterministic Markov base policy at every leaf. Fig. 2 gives a worked decision on an illustrative graph, separating motion admissibility from completion certification and showing the continuation retained after replanning.

D. Properties and Exact Oracle

The coverage result below records the standard diminishing-returns structure of weighted set coverage [34]. The subsequent analysis makes the SOOA state and audit assumptions explicit and specializes rollout feasibility and policy-improvement arguments [9], [10] to the required-target terminal predicate.

PROPOSITION 1 (Monotone submodularity of fixed-mask coverage) *For an observation set $S \subseteq \{1, \dots, M\}$, let $F(S) = \sum_{i=1}^N w_i \mathbf{1}\{i \in \cup_{j \in S} G_j\}$. If all $w_i \geq 0$, then F is normalized, monotone, and submodular.*

Proof:

Algorithm 1 Viability-preserving rollout ADP

Require: Candidate graph, shield $\mathcal{U}_s$, required set $\mathcal{K}$, budget H , depth d

- 1: $s \leftarrow (0, \mathbf{0}, \mathbf{0}, H)$; $\pi \leftarrow ()$ $\triangleright \pi$ stores the selected action sequence
- 2: **while** $\Gamma_{\mathcal{K}}(\mathbf{m}) = 0$ and $h > 0$ **do**
- 3: Enumerate every $a \in \mathcal{U}_s(s)$
- 4: **for** each admissible a **do**
- 5: $\hat{Q}_d(s, a) \leftarrow \ell_{ja} + \hat{V}_{d-1}(f(s, a))$
- 6: **end for**
- 7: Discard every action with $\hat{Q}_d(s, a) = +\infty$
- 8: **if** no finite action remains **then**
- 9: **return** no certified completion
- 10: **end if**
- 11: $a^* \leftarrow \arg \min_a \hat{Q}_d(s, a)$ with deterministic ties
- 12: Append a^* and its audit record to π
- 13: $s \leftarrow f(s, a^*)$
- 14: **end while**
- 15: **return** π if $\Gamma_{\mathcal{K}}(\mathbf{m}) = 1$; otherwise return no certified completion

Clearly $F(\emptyset) = 0$. If $A \subseteq B$, every target covered by A is covered by B , so nonnegative weights give $F(A) \leq F(B)$. For $a \notin B$, the marginal gain $F(A \cup \{a\}) - F(A)$ is the total weight in $G_a \setminus \cup_{j \in A} G_j$. Since $A \subseteq B$, this set contains $G_a \setminus \cup_{j \in B} G_j$, proving diminishing returns. ■

Remark 1 (Role of submodularity) Proposition 1 also holds for required-target gain by assigning unit weight to $\mathcal{K}$ and zero elsewhere. It does not justify removing zero-gain connecting actions or supply a greedy approximation ratio for a directed inspection tour. Both SOOA cost and the admissibility of the next observation depend on the current node.

PROPOSITION 2 (Markov sufficiency) *For a fixed deterministic candidate graph with fixed node masks, required set $\mathcal{K}$, source-dependent edge records, target weights, revisit rule, and horizon, the state in (11) is Markov for the finite graph decision problem.*

Proof:

The current node j_k identifies the source of every possible next edge; $\mathbf{m}_k$ and fixed $\mathcal{K}$ determine the remaining inspection requirements; b_k determines which observations remain selectable under the no-revisit rule; and h_k determines the remaining budget. By assumption, visibility masks and edge costs and audits are deterministic records indexed only by the current and destination nodes. Equations (12), (13), and (15) therefore depend on no earlier history once $(j_k, \mathbf{m}_k, b_k, h_k)$ is given. The proposition concerns binary prescribed-observation acceptance; partial dwell, multiple independent look directions, or time-dependent illumination would require additional state. ■

PROPOSITION 3 (Compositional preservation of stored audits) *Consider a sequence $(\mathbf{a}_{j_{k-1}j_k})_{k=1}^L$ whose first source is c_0 , whose idealized rest-to-rest endpoints match*

exactly, and whose every edge has $\chi_{j_{k-1}j_k} = 1$. Then every stored sample of the concatenated sequence satisfies the input, speed, and clearance inequalities, every edge endpoint satisfies its terminal conditions, and every stored passive-drift rollout has nonnegative margin when that audit is enabled.

Proof:

By the definition of χ_{ij} , all stated conditions hold on each accepted edge record. Exact endpoint matching identifies the terminal state of edge k with the source state of edge $k+1$, so the records concatenate without an unmodeled jump. The sample set of the complete trajectory is the finite union of the per-edge sample sets; therefore every member of that union retains the corresponding per-edge audit result. ■

Remark 2 (Meaning of audit preservation) The proposition is a finite-record composition result, not a continuous-time barrier certificate. It does not cover inter-sample clearance, endpoint perturbations beyond tolerance, closed-loop tracking error, or robustness to model and state uncertainty.

PROPOSITION 4 (Rollout viability) *For the deterministic finite graph, if $\hat{V}_d(s) < +\infty$, Algorithm 1 selects a shield-admissible action whose successor has a finite goal-reaching completion within the remaining budget. Replanning preserves this property until the goal is reached.*

Proof:

Finiteness in (18) provides a certificate consisting of shield-admissible prefix actions followed, unless the goal is reached earlier, by a base-policy completion. The selected first action is part of such a certificate, so its successor retains a goal-reaching sequence within the reduced budget. When the depth- d tree is rebuilt, the certificate tail remains admissible; if another exact prefix action is required, the next action of the certified base completion supplies it. Determinism preserves the recorded masks, costs, and audits. Induction over the decreasing action budget therefore proves goal attainment. ■

PROPOSITION 5 (Base-policy no-degradation) *If the adaptive base policy reaches the goal from s , then $\hat{V}_d(s)$ is finite and $\hat{V}_d(s) \leq \hat{V}_0(s)$ for every $d \geq 1$. For a fixed depth d , the realized graph cost of repeated replanning in Algorithm 1 is no greater than $\hat{V}_d(s)$ and therefore no greater than the base-policy cost.*

Proof:

The first d base-policy actions, truncated if the goal is reached earlier, followed by the remaining base completion form a feasible certificate in (18); hence $\hat{V}_d(s) \leq \hat{V}_0(s)$. Now take a minimizing depth- d certificate at a replanning state and let s' be the successor of its first action a . The certificate tail, extended when necessary by the next base-policy action, is feasible in the rebuilt depth- d tree at s' . Therefore $\ell_{ja} + \hat{V}_d(s') \leq \hat{V}_d(s)$. Summing this inequality over successive replans telescopes to a

realized cost no greater than the initial $\hat{V}_d(s)$. The result concerns the graph stage cost; reductions in physical Δv are evaluated separately. ■

For reduced-graph certification, exact recursion caches (13) for every reachable tuple (j, m, b, h) and enumerates every action in $\mathcal{U}_s(s)$ rather than a local reversal neighborhood.

THEOREM 1 (Optimality on an exhausted reduced SOOA graph) *For fixed finite nodes, deterministic visibility masks, deterministic audited edge costs, a required set $\mathcal{K}$, the no-revisit rule, and horizon H , exhaustive evaluation of (13) returns a minimum-cost task-completing shield-feasible sequence when one exists. If no such sequence exists, it returns value $+\infty$ and reports infeasibility.*

Proof:

Proceed by induction on the remaining budget h . At $h = 0$, (13) returns zero exactly for a goal state and $+\infty$ otherwise, which is correct. Assume the recursion is correct for all states with at most $h - 1$ remaining actions. At a state with h actions, every feasible sequence begins with exactly one $a \in \mathcal{U}_s(s)$ and then a feasible continuation from $f(s, a)$; by the induction hypothesis, the recursion assigns that continuation its minimum cost or $+\infty$ if none exists. Minimizing over every admissible first action therefore returns the minimum total cost, or $+\infty$ when all continuations are infeasible. The claim is limited to the exhausted sampled graph and does not extend to omitted camera poses or continuous-space trajectories. ■

With maximum branching factor $B_{\max}$, depth d , and at most H base-policy actions, direct decision evaluation costs $O(HB_{\max}^{d+1})$ in the worst case. Memoization reduces repeated evaluation when prefixes reach the same mask state. Fig. 3 and Table III quantify the offline cost-computation trade-off for $d = 1, \dots, 6$.

V. Experimental Protocol

A. Fixed Geometry and Required-Target Profiles

The primary study uses the corrected public ISS mesh distributed by the National Aeronautics and Space Administration (NASA) [35]. Its complete transformed geometry contains 247 525 triangles. Ninety equal-represented-area samples define the original surface denominator; 41 are observable from at least one of the 24 retained candidate observations. The directed archive contains 24 initial-to-observation and 24×23 inter-observation records, of which 264 pass every enabled gate. Every method receives the same graph and scenario-specific available-node set. A subsequent development pilot samples additional viewpoint graphs on the same geometry, as described in Section E.

Required-target profiles are selected deterministically from geometry before scenario evaluation. The mission manifest records their IDs, coordinates, selection rule, visibility masks, and checksums. These synthetic profiles distribute prescribed samples across the observable

TABLE II

Frozen synthetic required-target profiles. Three equal-width slabs span the mesh longest axis; each slab uses centroid-first, normalized farthest-point selection. IDs omit the common mesh prefix. The nine-item profile is primary

Profile	Items	Slab counts	Required sample IDs
6-target	6	2/2/2	00055, 00052, 00007, 00001, 00033, 00015
9-target	9	3/3/3	00055, 00052, 00059, 00007, 00001, 00079, 00033, 00015, 00019
12-target	12	4/4/4	00055, 00052, 00059, 00064, 00007, 00001, 00079, 00045, 00033, 00015, 00019, 00024

library; they are not an operational ISS maintenance checklist.

The required set remains fixed when a scenario removes observation nodes. Losing every available view of a required item makes the task impossible in the library. Otherwise, a valid route or exhausted exact search establishes feasibility status; cases without either certificate remain unresolved.

B. Dynamics, Costs, and Observation Accounting

Each rest-to-rest SOOA lasts 90 s and uses fixed-step fourth-order Runge–Kutta (RK4) propagation with $\Delta t = 3$ s. The HCW mean motion is 1.13137×10^{-3} rad/s, the acceleration limit is $u_{\max} = 0.060$ m/s², and the speed limit is $v_{\max} = 2.0$ m/s. The vehicle radius is 0.80 m, the full-mesh safety margin is 2 m, and terminal position and velocity tolerances are 0.5 m and 0.05 m/s. For the reported simulations, we set $\lambda_v = 5$, $\lambda_e = 0.18$, $\lambda_s = 120$, and $\lambda_a = 0.05$, weighting maneuver increment, terminal tracking error, clearance violation, and per-edge action cost, respectively, in (8). The action budget is 14. Required-only completion has no additional global area threshold.

Target-identity unions exclude duplicate credit. Reporting includes required-target completion $C^{\mathcal{K}}$, priority-weighted inspectable score C^{insp} , and unweighted fractions over the 41 observable and 90 original samples. Equal represented sample areas make the last fraction the sampled-area estimate; priority multipliers affect only the weighted score.

C. Development History and Fresh Confirmation

The first required-target campaign evaluated three fixed profiles on 36 validation, 150 test, and 90 shifted scenarios without feasibility rejection. The nine-item profile was designated primary. Depth-three ADP completed 37/50 test and 10/30 shifted tasks on that profile, whereas greedy construction and its sequence-improvement descendants completed none. Table IV retains these development results, which motivated a completion-capable comparator.

A fresh confirmation compares depth-three ADP with one-step ADP and local search initialized by a completed one-step ADP route. Methods, penalties, scenario identities, and implementation hashes were frozen before evaluating the new outcomes. The confirmation retains the nine prescribed targets and depth-three policy, with 12 validation, 50 test, and 30 shifted scenarios. Its seeds are disjoint from all development runs, and every generated case is retained.

Each scenario samples a node-removal probability uniformly from $[0, 0.10]$, then removes candidates independently using that probability. It multiplies priorities for approximately 6%–14% of targets by a factor in $[2, 4]$. Shifted scenarios use removal probabilities in $[0.15, 0.25]$, priority fractions of 12%–20%, and factors in $[5, 8]$. Priorities affect background scores and the coverage-only ablation, but not the required-target goal or stage cost. All scenarios perturb one frozen graph, and distinct seeds may yield identical routes.

After confirmation, the depth diagnostic was extended to $d = 1, \dots, 6$ on the same 12 validation scenarios. Every depth used the fixed nine-target requirement, graph, budget, and implementation, including zero-gain connecting actions. The extension retains all cases and leaves the primary test and shifted comparisons unchanged. Fig. 3 shows the resulting cost, runtime, and safety-screen workload.

Complete-route timing includes preliminary base-policy construction and excludes graph loading and serialization. Each scenario–depth pair was measured once in ascending-depth order using Python 3.14 on an Apple M4/macOS host. Scheduling and thermal effects were uncontrolled. Safety-screen counts exclude the separate preliminary base-policy call.

All six depths completed the same nine validation tasks; the other three lacked required visibility. Depth three reduced mean graph cost by 11.19% relative to depth two. It captured 84.11% of the observed reduction from depth one to the lowest mean cost among the six depths. Depth four improved cost by a further 2.74%, at 4.63 times the median planning time and 4.74 times the mean safety-screen count. Depths five and six produced no further cost or completion improvement, while median runtime reached 17.50 and 52.52 times the depth-three value.

The fixed depth-three setting therefore offers a computation–cost compromise; depth four provides lower cost when additional computation is acceptable.

D. Matched Baselines and Statistical Reporting

The confirmation includes the task-aware greedy incumbent, one-step ADP, ADP-seeded local search, and depth-three ADP under identical tasks, dynamics, safety gates, and action budgets. One-step ADP uses (18) with $d = 1$. Local search applies three audited best-improvement reversal passes to its completed route and retains the feasible seed if improvement fails. This com-

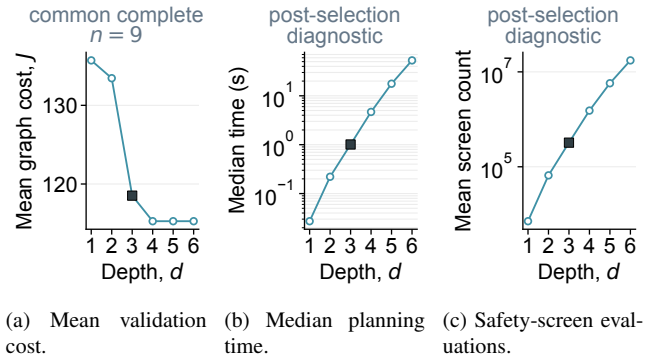


Fig. 3. Post-selection depth diagnostic. The square marks the primary depth-three setting. Cost uses the nine tasks completed at all depths; runtime and safety-screen counts include all 12 validation cases. Panels (b) and (c) use logarithmic scales. Table III reports completion counts and changes relative to depth three.

parator improves an ADP-initialized ordering; it does not introduce new viewpoints. Depth-three ADP uses the original greedy base value without a one-step fallback.

A coverage-only ablation retains the earlier 80% priority-weighted stopping rule and is evaluated against the same required-target identities. It tests the mission objective separately from the matched complete-task cost comparison.

Success and missing-target counts use every scenario; graph cost and physical Δv use jointly completed ADP/seeded-local pairs. For a route with final covered-target mask $\mathbf{m}$, the all-scenario score is $J_{\text{pen}} = J + 500(1 - C^{\mathcal{K}}(\mathbf{m})) + 500(1 - \Gamma_{\mathcal{K}}(\mathbf{m}))$, with fixed evaluation penalties distinct from physical maneuver cost. Paired-mean percentile intervals use 10 000 resamples, and exact two-sided sign tests exclude numerical ties. Timing includes initialization, including the one-step seed for local search; safety-screen counts cover the selected planner call.

E. All-Solvable Redundant-Viewpoint Benchmark

A subsequent pilot tested whether redundant viewpoints reduce the visibility bottleneck while permitting independent route certification. Three frozen seeds generated 33-view graphs on the same mesh, with the same nine required targets, initial state, camera model, dynamics, costs, and safety limits. Each graph contained three separated views per required target and six connecting views anchored to other observable samples. Candidate positions were sampled 26–42 m from their source target within a 35° cone around its surface normal, with the camera aimed at that target. From 64 attempts per source, deterministic first-valid and farthest-point selection retained geometry-valid views separated by at least 4 m. All rejected attempts were archived. Each graph’s $33 + 33 \times 32 = 1\,089$ directed transfers were regenerated and audited against the full mesh.

TABLE III

Post-selection required-target depth diagnostic on the same 12 validation scenarios. Graph-cost means use the 9 scenarios completed at all six depths. Time and screen counts include all 12 cases; ratios use depth three. Negative ΔJ denotes lower mean cost. All depths were rerun sequentially; the primary depth-three test results are unchanged

Depth	Complete	Mean J	ΔJ (%)	Median time (s)	Time ($\times$)	Mean screens	Screens ($\times$)
1	9/12	135.714	+14.51	0.0272	0.03	7,132	0.02
2	9/12	133.450	+12.60	0.2207	0.22	65,928	0.20
3	9/12	118.520	+0.00	1.0094	1.00	322,349	1.00
4	9/12	115.272	-2.74	4.6760	4.63	1,527,408	4.74
5	9/12	115.272	-2.74	17.6601	17.50	5,757,081	17.86
6	9/12	115.272	-2.74	53.0133	52.52	17,454,611	54.15

Each graph supplied one reference case, ten nominal cases and ten shifted cases, using the confirmation's node-removal probability ranges. All 63 cases were retained. The 59 cases with an independent feasible certificate define the primary all-solvable benchmark (three reference, 30 nominal and 26 shifted cases). The four remaining shifted cases are retained as a visibility-loss stress test and are excluded only from the primary solvable-case denominator. Background weights were fixed because they do not affect required-only completion or graph cost. The generation rules, seeds and source hashes were frozen before graph construction. This pilot followed inspection of the confirmation failures and is development evidence; it does not replace the frozen confirmation.

An independent mixed-integer linear program (MILP) selected a source-to-sink path over admissible edges. Binary arc and node variables enforced flow, required-target coverage, no revisits and the 14-observation budget; increasing node ranks excluded disconnected cycles. Its objective was the same additive graph cost, including the 0.05 action charge. A reconstructed valid route certified feasibility. Missing required visibility or a solver infeasibility result certified impossibility in the available graph; a limit without a valid route remained unresolved. Optimality was claimed only after solver proof and route-cost reconstruction. Certification preceded ADP evaluation and supplied no route to either ADP depth.

Greedy construction, one-step ADP, depth-three ADP and the independent MILP each received a nominal 15 s budget, including initialization or model construction and excluding shared graph generation. The pilot used Python 3.12.3 and SciPy 1.11.4/HiGHS on Ubuntu. Each method-case pair was timed once; greedy/ADP order was shuffled, while MILP timing came from the preceding certification phase. Native solver termination could slightly exceed its requested limit. The primary success-rate denominator is the independently certified all-solvable benchmark; all 63 generated cases and the four stress cases remain available for transparent audit. Cost gaps used only cases with proven optima.

TABLE IV

First-campaign development results by fixed profile. Local search in this campaign was initialized by the greedy incumbent, unlike the fresh confirmation comparator. Profiles use different seeds, so cross-profile rates are descriptive

Profile	Required items	Split	ADP complete	Local complete
6-target	6	Test	40/50	0/50
9-target	9	Test	37/50	0/50
12-target	12	Test	38/50	0/50
6-target	6	Shifted	10/30	0/30
9-target	9	Shifted	10/30	0/30
12-target	12	Shifted	10/30	1/30

VI. Results

A. Complete Required-Target Inspection

Depth-three ADP completed all nine targets in 43/50 test scenarios and 9/30 shifted scenarios. Seeded local search completed 43/50 and 8/30, respectively. Every completed mission achieved $C^K = 100\%$, with optional background coverage reported separately.

ADP completed every confirmation scenario retaining visibility of all required items. The seven incomplete test cases and 21 incomplete shifted cases each lost all available views of at least one mandatory target. Thus, ADP completed 43/43 and 9/9 feasible cases, respectively; their returned routes supplied feasibility witnesses. Five of the nine required targets originally had only one candidate view, making completion sensitive to node removal. These failures establish impossibility within the available library, not in continuous space. In the first development campaign, one six-target shifted case retained required visibility but produced no ADP certificate; its directed feasibility remains unresolved.

B. ADP Policy Improvement at Matched Completion

The predeclared ADP-minus-seeded-local penalized endpoint had mean differences of -12.256 on the 50 test scenarios and -31.999 on the 30 shifted scenarios. Their 95% intervals were $[-14.155, -10.212]$ and $[-91.494, -1.241]$, respectively. Physical-resource comparisons below use the jointly completed missions.

TABLE V

Required-target results. Resource means use successful routes of each method; the final column includes every scenario with the fixed failure penalty

Split	Method	Complete	Mean SOOAs	Mean Δv (m/s)	Mean J	Penalized J
Test	Task-aware incumbent	0/50	–	–	–	911.242
Test	One-step ADP	43/50	8.21	16.512	135.945	256.913
Test	ADP-seeded local search	43/50	8.21	15.777	129.892	251.707
Test	Depth-three ADP	43/50	9.00	14.033	115.641	239.451
Shifted	Task-aware incumbent	0/30	–	–	–	851.059
Shifted	One-step ADP	8/30	9.25	17.012	140.102	770.694
Shifted	ADP-seeded local search	8/30	8.88	15.888	130.841	768.224
Shifted	Depth-three ADP	9/30	9.11	14.656	120.751	736.225

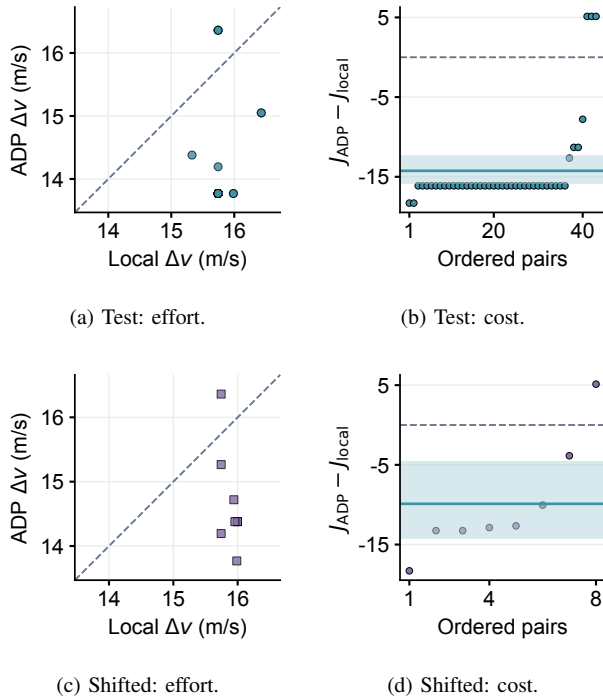


Fig. 4. ADP/seeded-local comparison on jointly completed missions: (a), (b) 43 test pairs; (c), (d) 8 shifted pairs. Panels (a), (c) compare physical Δv ; (b), (d) show ordered graph-cost differences in cost units. Dashed lines denote parity. Solid lines and bands give the means and 95% paired-bootstrap intervals: (b) -14.25 $[-15.72, -12.46]$; (d) -9.89 $[-14.11, -4.69]$.

Fig. 4 compares the 43 jointly completed test pairs and 8 shifted pairs. Mean ADP-minus-local graph-cost differences were -14.251 and -9.890 , respectively. Table VI reports the paired uncertainty and win/tie/loss counts.

On the 43 completed test pairs, mean Δv decreased from 15.777 m/s to 14.033 m/s, an 11.05% reduction. The paired reduction was 1.744 m/s, with a 95% interval of 1.525 – 1.923 m/s; ADP won 40 pairs and lost three. The eight shifted pairs had a mean reduction of 1.207 m/s. Its bootstrap interval excluded zero, while the exact sign test gave $p = 0.0703$.

The greedy base completed no confirmation mission, so this campaign measures improvement over one-step ADP and seeded local search rather than a completing

greedy comparator. Finite-graph unit tests separately exercise the base-cost bound, connecting actions, and exact recursion.

C. Objective Ablation and Safety

Fig. 5 summarizes component performance and the accepted-edge safety metrics. The coverage-only ablation in Table VII applies the fixed required-target denominator to routes using the 80% background stopping rule.

The coverage-only policy completed the nine-item requirement in 1/50 test and 0/30 shifted scenarios. In contrast, 41 of 43 completed ADP test missions finished below 80% weighted background coverage while accepting all nine prescribed items. Mean background coverage over those 43 successes was 72.32% .

Full-mesh occlusion and collision checks include structure outside the required target set. The reported clearance and input metrics describe the accepted deterministic edge records.

D. Representative Required-Target Route

Fig. 6 shows the selected routes, target completion, and accumulated Δv for the representative test pair.

The representative test case is the jointly completed scenario nearest the median ADP-minus-local graph-cost difference. Ties are resolved deterministically using the scenario identifier. Both routes complete all 9 prescribed targets. Rollout ADP uses 9 SOOAs, $J = 113.472$, and $\Delta v = 13.769$ m/s; local search uses 8 SOOAs, $J = 129.605$, and $\Delta v = 15.744$ m/s. The respective background inspectable scores are 77.22% and 64.97% . Repropagation reproduces both archived costs and velocity increments to numerical tolerance.

E. All-Solvable Benchmark Results

The three new graphs supplied at least three valid views of every required target, with 413, 439 and 386 admissible directed edges. Graph construction took 28.7–30.2 s per graph. Independent certification found all three reference cases, all 30 nominal cases and 26 of 30 shifted cases feasible. The other four shifted cases lacked

TABLE VI

ADP-minus-seeded-local effects. The predeclared penalized endpoint includes all scenarios; physical resources and graph cost use jointly complete pairs. Intervals are percentile 95% paired-bootstrap intervals; sign tests exclude ties

Split	Endpoint	Pairs	Mean difference	95% interval	Win/tie/loss	Sign-test p
Test	J_{pen} (all)	50	-12.256	$[-14.155, -10.212]$	40/7/3	3.02×10^{-9}
Test	J	43	-14.251	$[-15.715, -12.461]$	40/0/3	3.02×10^{-9}
Test	Δv (m/s)	43	-1.744	$[-1.923, -1.525]$	40/0/3	3.02×10^{-9}
Shifted	J_{pen} (all)	30	-31.999	$[-91.494, -1.241]$	8/21/1	0.0391
Shifted	J	8	-9.890	$[-14.111, -4.688]$	7/0/1	0.0703
Shifted	Δv (m/s)	8	-1.207	$[-1.713, -0.575]$	7/0/1	0.0703

TABLE VII

Required-target completion under the coverage-only objective. The required set is identical to that of the main comparison

Split	Objective ablation	All required complete
Test	Coverage-only (80%)	1/50
Shifted	Coverage-only (80%)	0/30

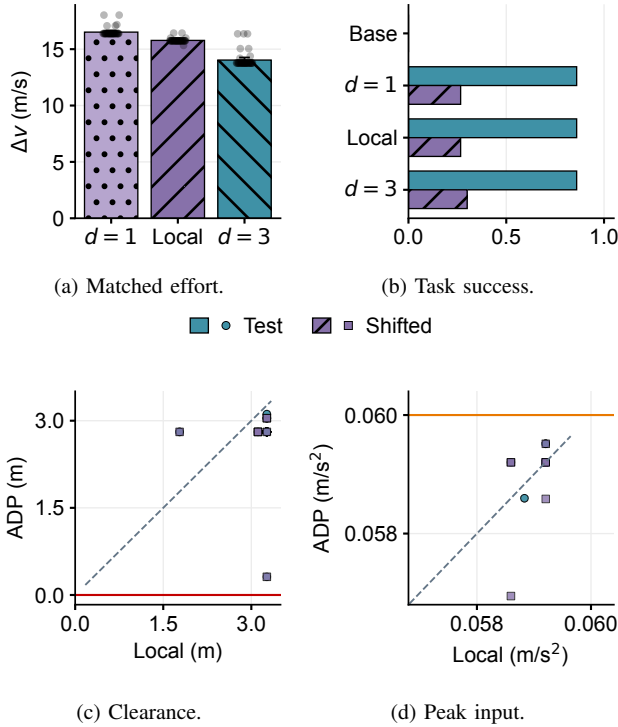


Fig. 5. Nine-target component and safety comparison. Base denotes the task-aware incumbent; $d = 1, 3$ denote ADP depths; Local denotes ADP-seeded local search. (a) Physical Δv : means and 95% bootstrap intervals over the 43 test cases completed by all three methods. (b) All-required success over all 50 test and 30 shifted cases. (c) Minimum clearance above the safety margin. (d) Peak commanded input; the orange line marks the 0.060 m/s^2 limit. Panels (c), (d) use jointly completed ADP/local pairs; dashed lines denote parity.

any available view of a required target; none remained unresolved. We therefore use the 59 certified-solvable cases as the primary benchmark for planner success rates

— Rollout ADP ♦ Required samples
— Seeded local ★ Initial state

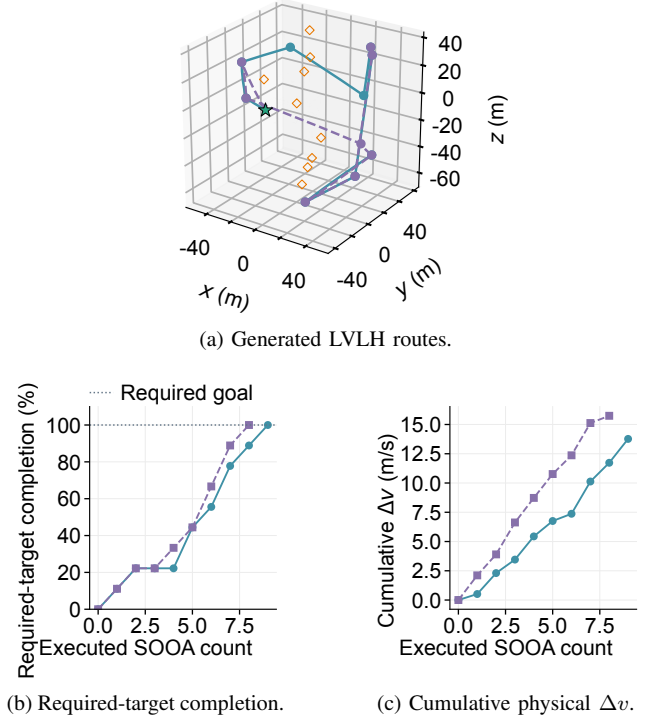


Fig. 6. Representative required-target test pair selected by the median-effect rule. Both methods complete the same required set. Markers identify terminal observations and prescribed samples; the star denotes the common initial state.

and retain the four cases in a separate visibility-loss stress test.

Table VIII reports completion and timing. Depth-three ADP completed all 59 feasible cases, whereas one-step ADP missed two feasible shifted cases. Greedy construction completed 13 nominal and 19 shifted cases, allowing direct checks of the conditional base-cost bound. Both ADP depths stayed within the cost of every completing greedy route, including the single completed reference, giving 66 method-case checks.

The MILP proved minimum graph costs for nine nominal and 18 shifted cases. Depth-three ADP's mean relative gaps were 0.86% and 1.01%, respectively, or 0.96% across these 27 cases. The other 32 feasible MILP runs in the all-solvable benchmark reached the time limit with checked routes whose optimality remained unproven.

TABLE VIII

Primary all-solvable benchmark from the development pilot. Each denominator is independently certified feasible: three reference, 30 nominal, 26 shifted and 59 cases overall. The four shifted cases failing the visibility certificate are retained in a separate stress test. Median planning times (nominal/shifted) are 0.001/0.001 s for the incumbent, 0.046/0.037 s for one-step ADP, 4.173/2.085 s for depth-three ADP and 15.007/5.373 s for the MILP. The MILP row is the inclusion certificate, not an independent planner baseline.

Method	Reference (3)	Nominal (30)	Shifted (26)	All-solvable (59)
Task-aware incumbent	1/3	13/30	19/26	33/59
One-step ADP	3/3	30/30	24/26	57/59
Depth-three ADP	3/3	30/30	26/26	59/59
Independent MILP	3/3	30/30	26/26	59/59

No ADP run timed out. The MILP returned slightly lower mean costs than depth-three ADP on the jointly completed nominal and shifted sets, by 0.150 and 1.388 cost units, respectively. Depth-three ADP reduced mean cost relative to one-step ADP by 5.493 units across 30 nominal pairs and 7.012 units across 24 shifted pairs, with longer computation times. These comparisons use explicit paired completion sets and do not establish uniform superiority over either comparator.

All 252 method–case records and 3 267 numeric edge records passed post-run reconstruction of coverage, route cost and enabled edge predicates. All graphs share one structure and initial state, and the nominal splits contain only seven, nine and ten distinct availability masks per ten cases. The all-solvable benchmark is therefore a feasibility-conditioned planner comparison, not a random unconditional sample. These dependent development observations warrant a larger confirmation with fresh seeds; they are not additional held-out evidence for the original 11.05% saving.

F. Required-Target ROS Execution

ROS-native HCW dynamics and linear-quadratic regulator (LQR) control track an ADP-derived reference fixed before execution, without online replanning. Each observation contributes its fixed geometric visibility mask only when position error is at most 0.5 m and speed is at most 0.05 m/s. Rejected views add no credit; repeated IDs count once. Execution success requires every observation to pass and every required target to be accepted; the supplemental hybrid case also requires 95% weighted background coverage. Both recorded event topics independently verify accepted-target unions.

The parent nine-target task imposes no background-coverage threshold; the supplemental case adds this threshold while retaining its target IDs and weights. Execution used 20-Hz control and state publication, 0.05 s integration steps, and a 0.060 m/s^2 acceleration limit. The filter retained a 1.50 m/s speed limit, 0.80 m vehicle radius, 2 m safety margin, and 8 m caution distance.

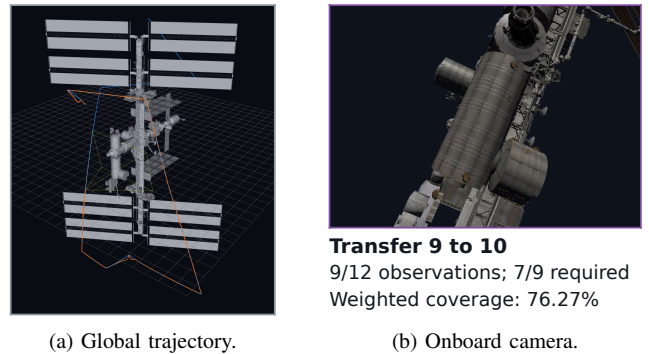
To increase the earlier nine-view result from 77.22% background coverage, we added a 95% goal using the same nine required IDs and 41-sample weights, and selected a twelve-view hybrid ADP plan. Synchronous screening rejected views 5 and 12 because their end-points lay inside the unchanged caution zone. We shifted them by 8 m and 6 m, adjusted their camera aims, and recomputed all transfers and visibility masks. The observation order, camera criteria, safety settings and acceptance tolerances were retained, with 60 s settling after each 90 s transfer. This post-hoc case, its failed diagnostic and refinement search are archived separately from the unchanged held-out study.

The subsequent normal-speed graphical run used Ubuntu 24.04.4, ROS 2 Jazzy, Gazebo Harmonic, RViz2, and Python 3.12.3. All twelve observations passed at 1800.035 s, accepting 9/9 required targets and 95.86% weighted background coverage (39/41 inspectable samples). Maximum terminal position error and speed were 0.050 m and 0.0038 m/s.

Integrating 36 172 timestamped safe-control messages through the final event gave 20.477 m/s executed Δv , versus 18.027 m/s for the transfer-and-hold reference. The safety stream reported 1 847 interventions. Auditing 36 181 trajectory samples and their connecting segments against 247 525 transformed triangles found no surface crossing. Minimum swept-body clearance was 3.380 m above the required 2 m margin; peak acceleration was 0.0600 m/s^2 . All execution and full-mesh audit gates passed. Reference-count completion passed, although the publisher reported a maximum 0.249 s gap, exceeding the 0.075 s nominal timing diagnostic.

Fig. 7 matches camera frames within 0.074 s of accepted event receipts. Both bags contain identical verification events, and named Gazebo poses confirm image–trajectory alignment. Images do not supply fresh visibility credit or defect detection. This selected, adjusted execution provides no paired ROS saving; the 11.05% result remains the unchanged offline aggregate.

Figure 8 shows a separate repeat execution of the same reference (Supplementary Video 1).



(a) Global trajectory.

(b) Onboard camera.

Fig. 8. RViz display: (a) planned (blue) and executed (orange) trajectories with camera FOV (yellow); (b) onboard Gazebo camera. Crops from one frame during transfer 9–10.

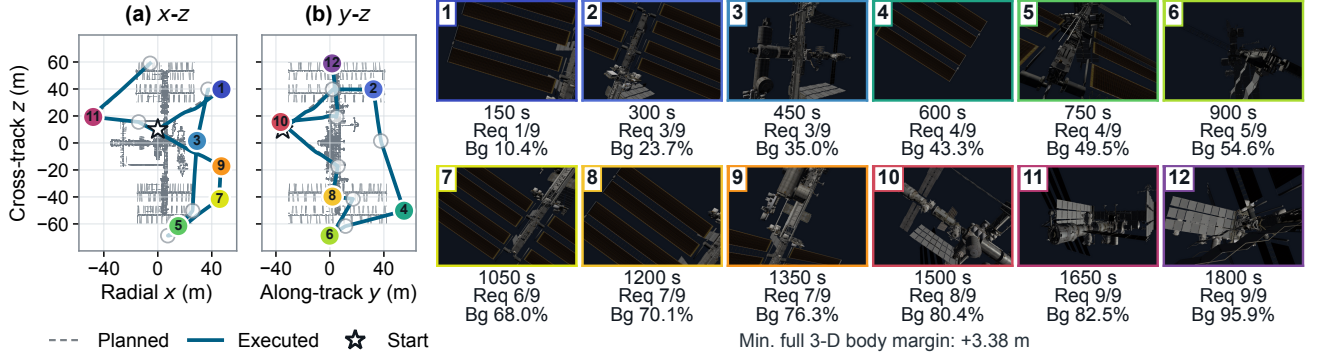


Fig. 7. Supplemental ROS 2 execution with twelve 90 s transfers and 60 s settling intervals. All observations pass, accepting 9/9 required targets and 95.86% weighted background coverage. Panels (a) and (b) highlight odd- and even-numbered views; hollow markers show the others. Both projections have equal spatial scale. Camera rows 1–6 and 7–12 show mission time, accepted required targets (Req), and weighted coverage (Bg). Uncropped frames and trajectories come from the same execution. Target credit uses fixed visibility masks and terminal tracking; the swept-body margin is audited against the full 3-D mesh.

VII. Discussion

The results support model-based rollout ADP for prescribed-target inspection. A base-policy completion supplies a checkable continuation, while Bellman lookahead improves the next action. The required-target constraint determines mission completion independently of optional background coverage. This sequencing layer complements learning-based proximity controllers [25], [27] and run-time-assured DRL inspection [29]; the reported baseline comparisons evaluate sequencing performance, not superiority over those controllers.

The task formulation also accommodates a hybrid goal with an additional background threshold, provided the base policy and rollout boundary use the same predicate. Target priorities alone cannot impose mandatory observations. The coverage-only ablation illustrates this difference directly.

The primary comparison measures maneuver savings against one-step ADP and its seeded reversal-search baseline. The viewpoint pilot additionally supplies independent feasibility certificates and proven cost optima for a subset of cases. Its common time budget exposes a trade-off: depth-three ADP improves completion and mean cost relative to one-step ADP, while MILP returns slightly lower mean costs with longer median runtimes. Repeated timing, broader search neighborhoods and a larger frozen confirmation are needed to assess this trade-off beyond the pilot.

Viewpoint availability is a separate requirement from successful sequencing. The original low shifted completion rate arose from missing mandatory visibility, so additional lookahead could not recover those cases. The redundant-viewpoint pilot addresses this observed limitation without changing required target identities. Its higher feasible fraction supports further evaluation of viewpoint redundancy, although the new library and protocol prevent interpreting the two campaigns as a paired algorithm improvement.

A. Scope and Limitations

The primary evidence covers synthetic prescribed samples on one deterministic observation graph; the development pilot adds three sampled graphs on the same mesh. Completion certifies those samples under the camera model, rather than full-region imagery or defect detection. Independent MILP certification checks route optimization, while visibility and clearance retain the existing geometry implementation. Operational inspection requires verified component requirements, denser sampling, resolution and dwell acceptance, illumination, and independent geometry.

The finite-completion safeguard may fail to find a route when its base policy is unsuccessful, even if another feasible route exists. The base-cost theorem applies when the base policy completes. Exact recursion and proven MILP optima concern only the specified finite graph; a time-limited feasible MILP route is not an optimum. Timing and confidence intervals characterize the stated desktop implementation and scenario generator, with no cross-host timing comparison. The observed depth-four-to-six plateau does not establish global optimality, and the pilot's 0.96% gap applies only to its 27 cases with proven optima.

The model assumptions in Table I exclude navigation uncertainty, actuator and attitude dynamics, and illumination effects. Safety remains conditional on the enabled deterministic audits, with passive-drift auditing disabled. The study evaluates model-based ADP; extending it to a learned critic or actor would require preserving a valid continuation and the relevant cost bound.

VIII. Conclusion

This paper developed viability-preserving rollout ADP for prescribed-target inspection over audited orbital arcs. A finite Bellman prefix and task-aware base-policy value

preserve completion and provide a conditional graph-cost bound.

ADP completed 43/50 test and 9/30 shifted missions, covering every feasible case in that confirmation; the remaining cases lacked required visibility. On jointly completed test missions, it reduced mean Δv by 11.05% relative to seeded local search. A subsequent redundant-viewpoint development pilot achieved 30/30 nominal and 26/30 shifted completions, plus all three reference cases, matching independently certified feasibility. Its mean graph-cost gap was 0.96% across 27 proven optima. These findings support completion-certified sequencing and motivate a larger, separately frozen evaluation of viewpoint redundancy on the prescribed inspection model.

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